

SWARAJ KUMAR SAHOO

1st August, 2004

Jajpur Road, Jajpur, Odisha

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Education

National Institute of Technology, Rourkela

INTEGRATED MSc MATHEMATICS (CGPA: 8.06)

Nov 2022 – Present

Rourkela, Odisha

Relevant Coursework

- Data Structures & Algorithm
- Database Management
- Object Oriented Programming
- Machine learning
- Deep learning
- Probability & Statistics

Projects

Accident-Gis

May 2026

Machine learning, APIs, langchain, React

GitHub

- * Developed a real-time AI-powered accident risk prediction platform using machine learning, GIS visualization, and live weather APIs.
- * Integrated browser geolocation and live weather data to generate dynamic accident risk predictions based on environmental conditions.
- * Implemented ML inference APIs using Scikit-learn and deployed predictive models for real-time severity estimation.
- * Designed GIS-based heatmaps and interactive dashboards to visualize accident-prone regions and AI-generated safety insights.
- * Built an interactive full-stack web application using React, FastAPI, and Leaflet for geospatial risk analysis and visualization.

Credit Risk Analyzer

Mar 2025

Material UI, APIs, Node.js, Express

GitHub

- * Designed and implemented backend APIs using Node.js and Express to ingest, validate, and persist multi-factor loan applicant data into a structured database.
- * Engineered data pipelines to process and aggregate 50+ applicant records across credit history, income, and behavioral factors for reliable risk assessment.
- * Integrated ML-driven risk scores into the application workflow, enabling lenders to make faster, data-backed lending decisions.

Achievements & Certifications

- **April Cohort Hackathon (Apr 2025):** Winner among 50+ teams for an industry-focused product showcased to 5+ mentors.
- **SBI ideathon (Feb 2025):** Secured Top 1000 out of 50000+ teams SBI IDEATHON for project *Self-Improvement*.
- **Hack4Bengal 4.0 (2025):** Ranked in Top 100 of 1500+ teams for "Hazard Reporter," improving citizen reporting by 50%.

Technical Skills

- **Languages:** Python, C, C++, HTML/CSS, JavaScript
- **Operating Systems:** Windows, MacOS
- **Developer Tools:** VS Code, Sublime, AWS
- **Frameworks:** React.js, Node.js, Express.js, Mongoose, Tailwind, Bootstrap, Redux

Extracurricular Activities

AXIOM, Math Club – NIT Rourkela

May 2024 – May 2025

Secretary

- * Led a 30-member team and organized 5+ campus-wide math events with 200+ participants.

Algorithmic & Programming Society – NIT Rourkela

Aug 2024 – May 2025

Event Management Lead

- * Managed AlgoVision for 200+ participants from 4+ departments ensuring zero logistical delays.